GUI-explorer: Autonomous Exploration and Mining of Transition-aware Knowledge for GUI Agent

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Paper version: arXiv 2505.16827v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *MLLMs misinterpret UI components and hold outdated knowledge.*

A **training-free** alternative to per-app fine-tuning: explore apps, mine what each action did, and retrieve it at test time with a ranker.

Motivation: three gaps

1. **Misread icons.** GPT-4o misidentifies a QQ Music button.
2. **App churn.** Amazon Shopping: 30 versions in 2024.
3. **Costly updates.** Per-app fine-tuning is too expensive.

Evaluation setup

SPA-Bench L3 single-app · AndroidWorld · 116 tasks · GPT-4o base model · SoM screenshot + a11y · SigLIP retrieval · branch 10 · depth 5 · 30-step limit

- **Online:** SPA-Bench, AndroidWorld. **Offline:** GUI-KRB, kept apart.
- **Exploration:** SPA-Bench guidance is mined in SPA-Bench.

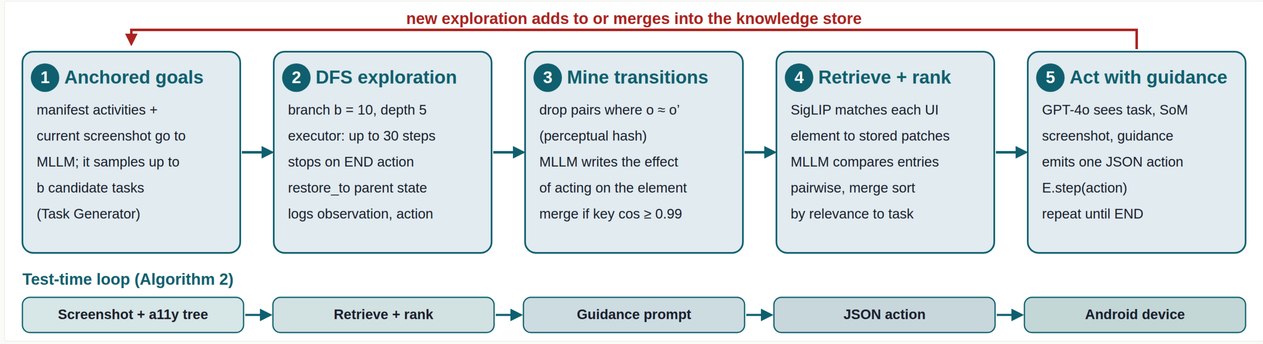
Our read (CUA Radar)

72	★★★	online
score of 100	band	evidence level

- **Reports:** online eval, baselines, cost, ablation, code.
- **Not reported:** seeds. **Flag:** AP5, exploration overlaps evaluation.
- Pattern C19. Authors' numbers, not rerun.

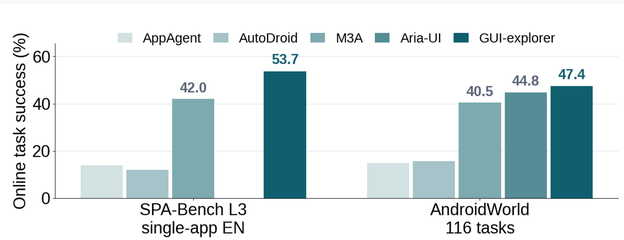
Method: explore, mine transitions, retrieve and rank guidance

- **Sees / acts:** SoM-annotated screenshot + a11y tree in; JSON action (click, scroll, input_text, open_app) out.
- **Knowledge:** each (screen, action, next screen) becomes a patch key and a text value; unchanged screens are dropped.
- **Signal:** manifest activities anchor the goals; only valid transitions are needed; no weights are updated.
- **Platform:** Android; manifest anchors limit it to mobile, web transfer shown only on 20 samples.



Our redraw (not the paper's figure): the exploration, mining and test-time guidance loop, from the method text.

Key results: online task success (%)

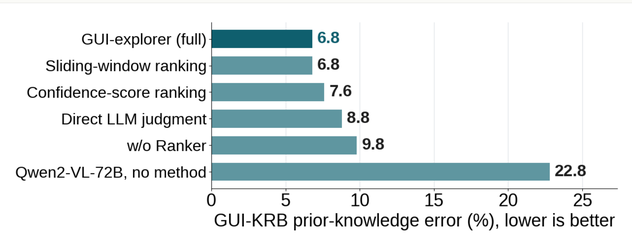


Tables 1 and 3, redrawn: SPA-Bench single-app English Level 3 and AndroidWorld.

53.7 vs M3A 42.0 · 47.4 vs Aria-UI 44.8

SO WHAT → *Baseline rows are taken from prior papers; no seeds or CIs reported. GUI-explorer uses SoM input with GPT-4o.*

Ablation: knowledge ranker (offline)



Tables 4 and 5, redrawn: error rate on the authors' GUI-KRB, Qwen2-VL-72B-Int4. Not task success.

SO WHAT → *Ranker off: 9.8 vs 6.8; sliding window ties at 6.8, but the paper argues it is sequential (O(n)) while merge sort parallelizes. On SPA-Bench, removing guidance costs 12.2% (Fig. 4). Offline metric, authors' own benchmark. Guidance mined in AndroidWorld still adds 4.3% on SPA-Bench.*

Headline Numbers

53.7%
SPA-Bench L3
success rate, GPT-4o base

47.4% AndroidWorld
1,300+ knowledge items
66 s per step

The paper's own controls

CONTROL	RESULT	REFERENCE
Remove dynamic guidance	12.2% drop, SPA-Bench	Fig. 4
Guidance from AndroidWorld	4.3% gain, SPA-Bench	vs no guidance
Screenshot-only goals	22.2 (2 AW apps)	16.7 none · 33.3 full
Cost / time per step	0.064 USD · 66.378 s	ranker: 28.462

No seeds or CIs reported

Takeaway

GPT-4o reaches 53.7% and 47.4%; guidance is mined in the evaluated apps.

no parameter updates · 1,300+ items · 46 apps · 66 s per step · RTX 4070 laptop (8GB)

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